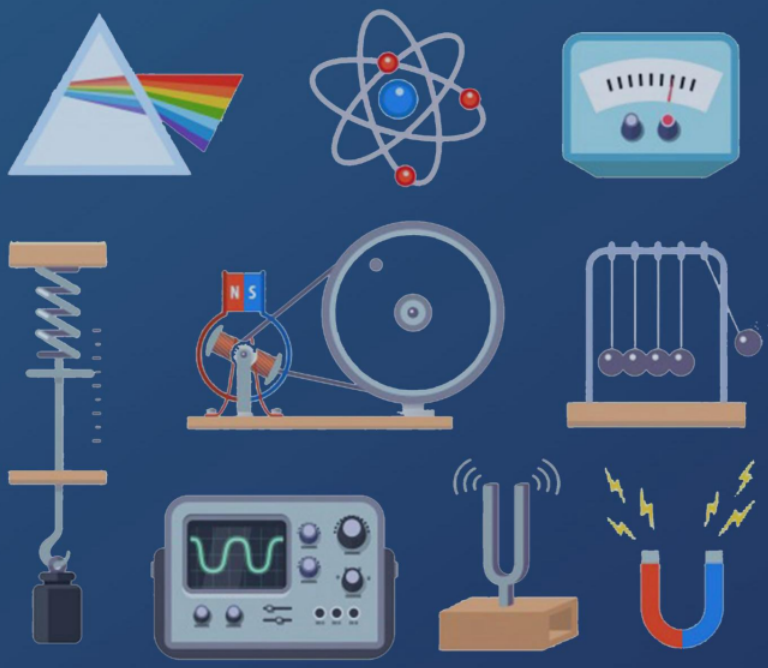
PHYSICS

Wave Optics



SHEET

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Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 4
5.	Exercise 2	Page - 6
6.	Exercise 2A	Page - 9
7.	Answer Key	Page - 14

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Wave particle duality	1			
2	Wave recape				
3	Huygen's principle				1
4	Interference of waves	2-7	1	1,2	
5	YDSE	8-13	2,3,5-7, 11,12	3-9	2-7
6	Variation in intensity	14,15		10	8
7	Shape of fringes				
8	YDSE with white light	16	4	11	
9	Variation in ydse			12,13	9
10	Optical path	17-23	8-10	14-19	10-12
11	Diffraction	24-26		20,21	
12	Polarisation	27-30		23,23	
13	After Full Chapter				13-24

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Wave particle duality

Q1. Through which character we can distinguish the light waves from sound waves :

(A) Interference (B) Refraction

(C) Polarisation (D) Reflection

Interference of waves

Q2. The phase difference corresponding to path difference of x is :

(A) $\frac{2\pi x}{\lambda}$ (B) $\frac{2\pi\lambda}{x}$

(C) $\frac{\pi x}{\lambda}$ (D) $\frac{\pi\lambda}{x}$

Q3. If intensity of each of the two waves is I and they are having phase difference of 120°, when the waves are superposed, then the resultant intensity will be :

(A) I (B) 2I

(C) I/2 (D) 4I

Q4. Two wave are represented by the equations $y_1 = a \sin \omega t$ and $y_2 = a \cos \omega t$. The first wave :-

(A) leads the second by π

(B) lags the seconds by π

(C) leads the second by $\frac{\pi}{2}$

(D) lags the seconds by $\frac{\pi}{2}$

Q5. The resultant amplitude of a vibrating particle by the superposition of the two waves $y_1 = a\sin\left[\omega t + \frac{\pi}{3}\right]$ and $y_2 = a \sin \omega t$ is :-

(A) a (B) $\sqrt{2}$ a

(C) 2a (D) $\sqrt{3}$ a

Q6. Phenomenon of interference is not observed by two sodium lamps of same power. It is because both waves have :-

(A) not constant phase difference

(B) zero phase difference

(C) different intensity

(D) different frequencies

Q7. Two coherent sources of intensities I_1 and I_2 produce an interference pattern. The maximum intensity in the intrference pattern will be :-

(A) $I_1 + I_2$ (B) $I_1^2 + I_2^2$

(C) $(I_1 + I_2)^2$ (D) $(\sqrt{I_1} + \sqrt{I_2})^2$

YDSE

Q8. In Young's experiment, if X_{mr} and X_{mv} denotes the distances of mth red and violet fringe from the central fringe. Then :

(A) $X_{mr} > X_{mv}$ (B) $X_{mr} < X_{mv}$

(C) $X_{mr} = X_{mv}$ (D) $X_{mr} + X_{mv} = 0$

Q9. In Young's double slit experiment, wave-length of light is 6000Å. Then the phase difference between the light waves reaching the third bright fringe from the central fringe will be :

(A) zero (B) 2π

(C) 4π (D) 6π

Q10. If in a Young's double slit experiment, width between the slits is 3 cm, the separation between slits and screen is 7 cm and wavelength of light is 1000 Å, then fringe width will be :

(A) 2×10^{-5} m. (B) 2×10^{-9} m

(C) 0.2×10^{-6} m (D) 2.3×10^{-7} m

Q11. Young's experiment establishes that :-

(A) light consists of waves

(B) light consists of particles

(C) light consists of neither particles nor waves

(D) light consists of both particles and waves

Q12. In YDSE experiment, when two light rays make third minima, then they have :-

(A) phase difference of 3π

(B) phase difference of $\frac{5\pi}{2}$

(C) path difference of 3λ

(D) path difference of $\frac{5\lambda}{2}$

Q13. In an interference pattern of two waves fringe width is β . If the frequency of source is doubled then fringe width will become :-

(A) $\frac{1}{2}\beta$ (B) β (C) 2β (D) $\frac{3}{2}\beta$

Variation in intensity

Q14. If ratio of amplitude of two interferring source is 3 : 5. Then ratio of intensity of maxima and minima in interference pattern will be :

(A) 25 : 16 (B) 5 : 3

(C) 16 : 1 (D) 25 : 9

Q15. In the young's double slit experiment the intensity of central maxima is observed to be I_0. If one of the slits is covered, then intensity at the central maxima will become :-

(A) $\frac{I_0}{2}$ (B) $\frac{I_0}{\sqrt{2}}$ (C) $\frac{I_0}{4}$ (D) I_0

YDSE with white light

Q16. In Young's double slit experiment, if monochromatic light is replaced by white light :-

(A) all bright fringes become white

(B) all bright fringes have coloures between violet and red

(C) only the central fringe is white, all other fringes are coloured

(D) no fringes are observed

Optical path

Q17. In Young's experiment, monochromatic light through a single slit S is used to illuminate the two slits S_1 and S_2. Interference fringes are obtained on a screen. The fringe width is found to be w. Now a thin sheet of mica (thickness t and refractive index μ) is placed near and in front of one of the two slits. Now the fringe width is found to be w′, then :

(A) $w' = w/\mu$ (B) $w' = w\mu$

(C) $w' = (\mu - 1)\,tw$ (D) $w' = w$

Q18. If a transparent medium of refractive index $\mu = 1.5$ and thickness $t = 2.5 \times 10^{-5}$ m is inserted in front of the slits of Young's Double slit experiment, how much will be the shift in the interference pattern? The distance between the slits is 0.5 mm and that between slits and screen is 100 cm :

(A) 5 cm (B) 2.5 cm

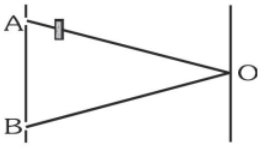
(C) 0.25 cm (D) 0.1 cm

Q19. InYoung's double slit experiment, a mica sheet of thickness t and refractive index μ is introduced in the path of ray from the first source S_1. By how much distance the fringe pattern will be displaced.

(A) $\frac{d}{D}(\mu - 1)t$ (B) $\frac{D}{d}(\mu - 1)t$

(C) $\frac{d}{(\mu - 1)D}$ (D) $\frac{D}{d}(\mu - 1)$

Q20. In Young's experiment, monochromatic light is used to illuminate the two slits A and B. Interference fringes are observed on a screen placed in front of the slits. Now if a thin glass plate is placed normally in the path of the beam coming from the slit then :



(A) the fringes will disappear
(B) the fringe width will decrease
(C) the fringe width will increase
(D) there will be no change in the fringe width

Q21. A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ. If μ represents the refractive index of glass with respect to air, then the angle between reflected and refracted rays is :

(A) $90 + \phi$ (B) $\sin^{-1}(\mu\cos\phi)$
(C) $90°$ (D) $90° - \sin^{-1}(\sin\phi/\mu)$

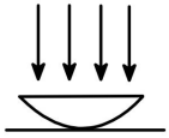
Q22. **Statement-1 :** In YDSE, as shown in figure, central bright fringe is formed at O. If a liquid is filled between plane of slits and screen, the central bright fringe is shifted in upward direction.

Statement-2 : If path difference at O increases, y-coordinate of central bright fringe will change.



(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Q23. A thin slice is cut out of a glass cylinder along a plane parallel to its axis. The slice is placed on a flat glass plate as shown. The observed interference fringes from this combination shall be :



(A) Straight
(B) Circular
(C) Equally spaced
(D) Having fringe spacing which increases as we go outwards.

Diffraction

Q24. A light source of 5000Å wave length produces a single slit diffraction. The first minima in diffraction pattern is seen, at a distance of 5mm from central maxima. The distance between screen and slit is 2 metre. The width of slit in mm will be :

(A) 0.1 (B) 0.4
(C) 0.2 (D) 2

Q25. A plane wave front of wave length 6000 Å is incident upon a slit of 0.2mm width, which enables Fraunhofer's diffraction pattern to be obtained on a screen 2 metre away. Width of the central maxima in mm will be :

(A) 10 (B) 12
(C) 8 (D) 2

Q26. Central fringe obtained in diffraction pattern due to a single slit :–

(A) is of minimum intensity
(B) is of maximum intensity
(C) intensity does not depend upon slit width
(D) none of the above

Polarisation

Q27. Refractive index of water is 1.33. Calculate the angle of polarisation for light reflected from the surface of a lake.

(A) 60° (B) 53°

(C) 37° (D) 45°

Q28. A parallel beam of light is incident at an angle of 60° on a plane glass surface and the reflected beam is completely polarised. What is the angle of refraction in glass ?

(A) 60° (B) 53°

(C) 37° (D) 30°

Q29. The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n).

(A) $\sin^{-1}(n)$ (B) $\sin^{-1}\left(\frac{1}{n}\right)$

(C) $\tan^{-1}\left(\frac{1}{n}\right)$ (D) $\tan^{-1}(n)$

Q30. A polaroid is placed at 45° to an incoming light of intensity I_0. Now the intensity of light passing through polaroid after polarisation would be :

(A) I_0 (B) $I_0/2$

(C) $I_0/4$ (D) Zero

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. To observe a sustained interference pattern formed by two light waves, it is not necessary that they must have :

(A) The same frequency

(B) Same amplitude

(C) A constant phase difference

(D) The same intensity

Q2. If the source of light used in a Young's Double Slit Experiment is changed from red to blue, then

(A) The fringes will become brighter

(B) Consecutive fringes will come closer

(C) The number of maxima formed on the screen increases

(D) The central bright fringe will become a dark fringe.

Q3. If one of the slits of a standard YDSE apparatus is covered by a thin parallel sided glass slab so that it transmit only one half of the light intensity of the other, then :

(A) The fringe pattern will get shifted towards the covered slit.

(B) The fringe pattern will get shifted away from the covered slit.

(C) The bright fringes will be less bright and the dark ones will be more bright.

(D) The fringe width will remain unchanged.

Q4. In a YDSE apparatus, if we use white light then :

(A) The fringe next to the central will be red

(B) The central fringe will be white.

(C) The fringe next to the central will be violet

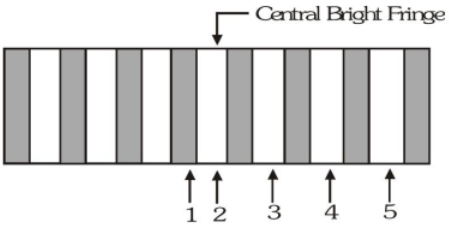
(D) There will not be a completely dark fringe.

PARAGRAPH TYPE

This section contains ***PARAGRAPHS****. Based on each paragraph, there are questions. Each question has FOUR options.* ***ONLY ONE*** *of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 5 to 7

The figure shows the interference pattern obtained in a double–slit experiment using light of wavelength 600nm.



Q5. The third order bright fringe is :

(A) 2 (B) 3 (C) 4 (D) 5

Q6. Which fringe results from a phase difference of 4π between the light waves incidenting from two slits?

(A) 2 (B) 3 (C) 4 (D) 5

Q7. Let ΔX_A and ΔX_C represent path differences between waves interfering at 1 and 3 respectively then $(|\Delta X_C| - (|\Delta X_A|)$ is equal to :

(A) 0 (B) 300 mn

(C) 600 nm (D) 900 nm

Paragraph for Question No. 8 to 10

The lens governing the behavior of the rays namely rectilinear propagation, laws of reflection and refraction can be summarised in one fundamental law known as Fermat's principle. According to this principle a ray of light travels from one point to another such that the time taken is at a stationary value (maximum or minimum). If c is the velocity of light in a vacuum, the velocity in a medium of refractive index μ is $\frac{c}{\mu}$, hence time taken to travel a distance ℓ is $\frac{\mu\ell}{c}$. If the light passes through a number of media, the total time taken is $\left(\frac{1}{c}\right)\sum\mu\ell$ or $\frac{1}{c}\int\mu d\ell$ if refractive index varies continuously.

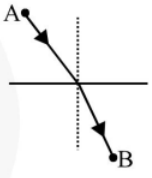
Now, $\sum\mu\ell$ is the total optical path, so that Fermat's principle states that the path of a ray is such that the optical path in at a stationary value. This principle is obviously in agreement with the fact that the ray are straight lines in a homogenous isotropic medium. It is found that it also agrees with the classical laws of reflection and refraction.

Q8. If refractive index of a slab varies as $\mu = 1 + x^2$ where x is measured from one end, then optical path length of a slab of thickness 1 m is :

(A) $\frac{4}{3}$m (B) $\frac{3}{4}$m

(C) 1 m (D) None

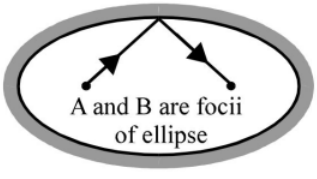
Q9. The optical path length followed by ray from point A to B given that laws of refraction are obeyed as shown in figure is :-



(A) Maximum (B) Minimum

(C) Constant (D) None

Q10. The optical path length followed by ray from point A to B given that laws of reflection are obeyed as shown in figure is :



(A) Maximum (B) Minimum

(C) Constant (D) None

SUBJECTIVE TYPE

Q11. Waves emitted by two identical sources produces intensity of K unit at a point on screen where path difference between these waves is λ, calculate the intensity at that point on screen at which path difference is $\frac{\lambda}{4}$.

Q12. In young's double slit experiment, two slits are separated by 3mm distance and illuminated by light of wavelength 480 nm. The screen is at 2m from the plane of the slits. Calculate the separation between the 8th Bright fringe and the 3rd dark fringe obtained on same side of central Bright fringe.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Two coherent sources of equal intensities produce a maximum of 100 units. If the amplitude of one of the sources is reduced by 20%, then the maximum intensity produced will be :

(A) 100 (B) 81

(C) 89 (D) 60

Q2. Two waves $Y_1 = a\sin\omega t$ and $Y_2 = a\sin(\omega t+\delta)$ are producing interference, then resultant intensity is :-

(A) $a^2\cos^2\left(\frac{\delta}{2}\right)$ (B) $2a^2\cos^2\left(\frac{\delta}{2}\right)$

(C) $3a^2\cos^2\left(\frac{\delta}{2}\right)$ (D) $4a^2\cos^2\left(\frac{\delta}{2}\right)$

Q3. In Young's double slit experiment, if the width of the slits are in the ratio 4 : 9 the ratio of the intensity of maxima to the intensity at minima will be :

(A) 169 : 25 (B) 81 : 16

(C) 25 : 1 (D) 9 : 4

Q4. The path difference between two interfering waves at a point on the screen is λ/6. The ratio of intensity at this point and that at the central bright fringe will be :

(Assume that intensity due to each slit is same)

(A) 0.853 (B) 8.53 (C) 0.75 (D) 7.5

Q5. In a young double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is changed to 400 nm, number of fringes observed in the same segment of the screen is given by :

(A) 12 (B) 18 (C) 24 (D) 30

Q6. In Young's double slit experiment, the wavelength of red light is 7800 Å and that of blue light is 5200Å. The value of n for which n^{th} bright band due to red light coincides with $(n+1)^{th}$ bright band due to blue light, is :

(A) 1 (B) 2 (C) 3 (D) 4

Q7. In Young's double slit experiment, the two slits act as coherent sources of equal amplitude A and wavelength λ. In another experiment with the same setup the two slits are sources of equal amplitude A and wavelength λ but are incoherent. The ratio of the average intensity of light at the midpoint of the screen in the first case to that in the second case is :-

(A) 1 : 1 (B) 2 : 1

(C) 4 : 1 (D) none of these

Q8. If intensity of each wave in the observed interference pattern in Young's double slit experiment is I_0, then for some point P where the phase difference is ϕ , intensity I will be:

(A) $I = I_0 \cos\phi$ (B) $I = I_0 \cos^2 \phi$

(C) $I = I_0 (1+\cos\phi)$ (D) $I = 2I_0(1+\cos\phi)$

Q9. In Young's double slit experiment, one of the slits is so painted that intensity of light emitted from it is half of that of the light emitted from other slit. Then :-

(A) Fringe system will disappear

(B) Bright fringes will become brighter and dark fringes will be darker

(C) Both bright and dark fringes will become darker

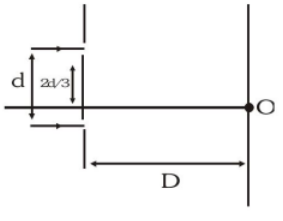
(D) Dark fringes will become less dark and bright fringes will become less bright.

Q10. Two beams of light having intensities I and 4I interfer to produce a fringe pattern on a screen. The phase difference between the beam is $\frac{\pi}{2}$ at point A and 2π at point B. Then find out the difference between the resultant intensities at A and B.

(A) 2I (B) 5I

(C) I (D) 4I

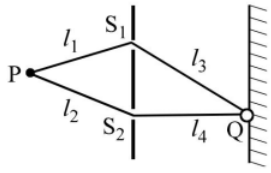
Q11. In the figure shown if a parallel beam of white light is incident on the plane of the slits then the distance of the nearest white spot on the screen from O is : [assume d << D, λ << d]



(A) 0 (B) d/2

(C) d/3 (D) d/6

Q12. Two identical narrow slits S_1 and S_2 are illuminated by light of wavelength λ from a point source P. If, as shown in the diagram, the light is then allowed to fall on a screen, and if n is a positive integer, the condition for destructive interference at Q is :-



(A) $(\ell_1 - \ell_2) = (2n + 1)\lambda/2$

(B) $(\ell_3 - \ell_4) = (2n + 1)\lambda/2$

(C) $(\ell_1 + \ell_2) - (\ell_3 + \ell_4) = n\lambda$

(D) $(\ell_1 + \ell_3) - (\ell_2 + \ell_4) = (2n + 1)\lambda/2$

Q13. In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits so that it is closer to the upper slit, as shown. Then :-



(A) The fringe width will increase and fringe pattern will shift down.

(B) The fringe width will remain same but fringe pattern will shift up.

(C) The fringe width will decrease and fringe pattern will shift down.

(D) The fringe width will remain same but fringe pattern will shift down.

Q14. In Young's experiment, light of wavelength 6000Å is used to produce fringes of width 0.8 mm at a distance of 2.5 m. If the whole experiment is deep in a liquid of refractive index 1.6, then fringe width will be :

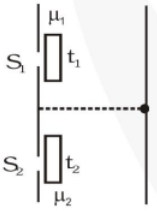
(A) 0.5 mm (B) 0.6 mm

(C) 0.4 mm (D) 0.2 mm

Q15. The central fringe of interference pattern produced by light of wavelength 6000Å is found to shift to the position of 4th bright fringe, after a glass plate of $\mu = 1.5$ is introduced. The thickness of the glass plate is :

(A) 4.8 μm (B) 8.23 μm
(C) 14.98 μm (D) 3.78 μm

Q16. In a Young's double slit experiment, a slab of thickness 1.2 μm and refractive index 1.5 is placed in front of one slit and another slab of thickness t and refractive index 2.5 is placed in front of the second slit. If the position of the central fringe remains unaltered, then the thickness t is :-



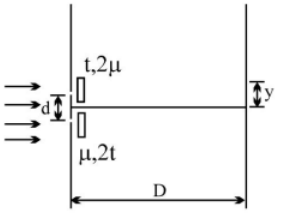
(A) 0.4 μm (B) 0.8 μm
(C) 1.2 μm (D) 7 μm

Q17. In the ideal double–slit experiment, when a glass–plate (refractive index 1.5) of thickness t is introduced in the path of one of the interfering beams (wavelength λ), the intensity at the position

where the central maximum occurred previously remains unchanged. The minimum thickness of the glass–plate is :-

(A) 2λ (B) $\frac{2\lambda}{3}$ (C) $\frac{\lambda}{3}$ (D) λ

Q18. In the YDSE shown the two slits are covered with thin sheets having thickness t & $2t$ and refractive index 2μ and μ. Find the position (y) of central maxima.



(A) zero (B) $\frac{tD}{d}$

(C) $-\frac{tD}{d}$ (D) None

Q19. A thin film of thickness t and index of refraction 1.33 coats a glass with index of refraction 1.50. What is the least thickness t that will strongly reflect light with wavelength 600 nm incident normally?

(A) 226 nm (B) 300 nm
(C) 400 nm (D) 450 nm

Q20. In a single slit diffraction experiment first minimum for $\lambda_1 = 660$ nm coincides with first maxima for wavelength λ_2. Calculate λ_2.

(A) 550 nm (B) 400 nm
(C) 440 nm (D) 660 nm

Q21. Find the half angular width of the central bright maximum in the Fraunhofer diffraction pattern of a slit of width 12×10^{-5} cm, when the slit is illuminated by monochromatic light of wavelength 6000 Å.

(A) 40° (B) 45°
(C) 30° (D) 60°

Q22. The angle of polarisation for any medium is 60°, what will be critical angle for this :

(A) $\sin^{-1}\sqrt{3}$ (B) $\tan^{-1}\sqrt{3}$

(C) $\cos^{-1}\sqrt{3}$ (D) $\sin^{-1}\frac{1}{\sqrt{3}}$

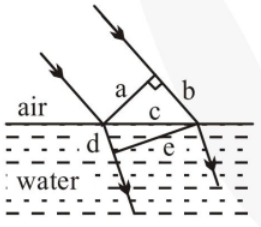
Q23. Two Nicols are oriented with their principal planes making an angle of 60°. The percentage of intansity of incident unpolarized light which passes through the system is :

(A) 50% (B) 100%
(C) 12.5% (D) 37.5%

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Figure shows plane waves refracted from air to water using Huygen's principle (where a, b, c, d, e are lengths on the diagram). The refractive index of water wrt air is the ratio:-



(A) a/e (B) b/e
(C) b/d (D) d/b

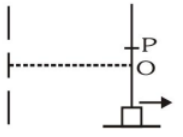
Q2. If the first minima in a Young's slit experiment occurs directly in front of one of the slits, (distance between slit & screen D = 12 cm and distance between slits d = 5 cm) then the wavelength of the radiation used can be :

(A) 2 cm (B) 4 cm
(C) $\frac{2}{3}$ cm (D) $\frac{4}{3}$ cm

Q3. In young's double slit experiment, slits are arranged in such a way that besides central bright fringe, there is only one bright fringe on either side of it. Slit separation d for the given condition cannot be (if λ is wavelength of the light used) :

(A) λ (B) $\lambda/2$
(C) 2λ (D) $3\lambda/2$

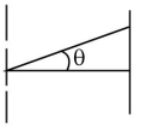
Q4. In a Young's Double slit experiment, first maxima is observed at a fixed point P on the screen. Now the screen is continuously moved away from the plane of slits. The ratio of intensity at point P to the intensity at point O (centre of the screen) :-



(A) Remains constant
(B) Keeps on decreasing
(C) First decreases and then increases
(D) First decreases and then becomes constant

Q5. Two slits are separated by 0.3 mm. A beam of 500 nm light strikes the slits producing an interference pattern. The number of maxima observed in the angular range $-30° < \theta < 30°$.

(A) 300
(B) 150
(C) 599
(D) 601



Q6. In a YDSE bi–chromatic light of wavelengths 400nm and 560 nm are used. The distance between the slits is 0.1 mm and the distance between the plane of the slits and the screen is 1m. The minimum distance between two successive regions of complete darkness is :–

(A) 4 mm (B) 5.6 mm
(C) 14 mm (D) 28 mm

Q7. In Young's double slit experiment intensity at a point is (1/4) of the maximum intensity. Angular position of this point is :–

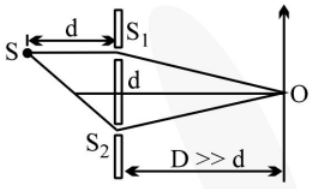
(A) $\sin^{-1}\left(\frac{\lambda}{d}\right)$ (B) $\sin^{-1}\left(\frac{\lambda}{2d}\right)$
(C) $\sin^{-1}\left(\frac{\lambda}{3d}\right)$ (D) $\sin^{-1}\left(\frac{\lambda}{4d}\right)$

Q8. In a Young's double slit experiment D equals the distance of screen and d is the separation between the slit. The distance of the nearest point to the central maximum where the intensity is same as that due to a single slit, is equal to :-

(A) $\frac{D\lambda}{d}$ (B) $\frac{D\lambda}{2d}$

(C) $\frac{D\lambda}{3d}$ (D) $\frac{2D\lambda}{d}$

Q9. To make the central fringe at the centre O, a mica sheet of refractive index 1.5 is introduced. Choose the correct statements (s).



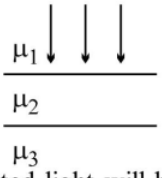
(A) The thickness of sheet is $2(\sqrt{2}-1)d$ in front of S_1.

(B) The thickness of sheet is $(\sqrt{2}-1)d$ in front of S_2.

(C) The thickness of sheet is $2\sqrt{2}\,d$ in front of S_1.

(D) The thickness of sheet is $(2\sqrt{2}-1)d$ in front of S_1.

Q10. Consider a case of thin film interference as shown. Thickness of film is equal to wave-length of light in μ_2.



(A) Reflected light will be maxima if $\mu_1 < \mu_2 < \mu_3$

(B) Reflected light will be maxima if $\mu_1 < \mu_2 > \mu_3$

(C) Transmitted light will be maxima if $\mu_1 > \mu_2 > \mu_3$

(D) Transmitted light will be maxima if $\mu_1 > \mu_2 < \mu_3$

Q11. In a Young's double-slit experiment, let A and B be the two slits. Films of thicknesses t_A and t_B and refractive indices μ_A and μ_B, are placed in front of A and B respectively.

If $\mu_A t_A = \mu_B t_B$, the central maximum will :

(A) not shift

(B) shift towards A

(C) shift towards B

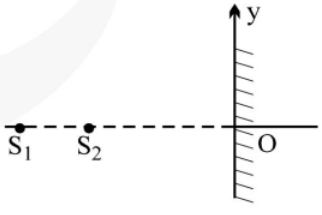
(D) option (B), if $t_B > t_A$; option (C) if $t_B < t_A$

Q12. It is necessary to coat a glass lens with a non-reflecting layer. If the wavelength of the light in the coating is λ, the best choice is a layer of material having an index of refraction between those of glass and air and a minimum thickness of :-

(A) $\frac{\lambda}{4}$ (B) $\frac{\lambda}{2}$

(C) $\frac{3\lambda}{8}$ (D) λ

Q13. Two point monochromatic and coherent sources of light of wavelength λ are placed on the dotted line in front of a large screen. The sources emit waves in phase with each other. The distance between S_1 and S_2 is d while their distance from the screen is much larger. Which is the set of correct statement:



(A) If d = 7 λ/2, O will be a minima

(B) If d = 4.3 λ, there will be a total of 8 minima on y-axis.

(C) If d = 7 λ, O will be a maxima.

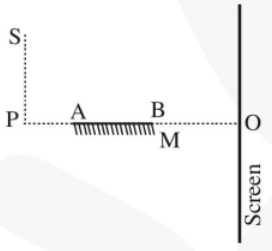
(D) If d = λ, there will be only one maxima on the screen.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 14 and 15

In an experiment on interference due to single mirror, a light wave emitted directly by the source S (narrow slit) interferes with the wave reflected from the mirror M of length 2 mm. Source and screen are separated by distance 90 cm. Source *S* is at the height of 3 mm, from the point P and the middle point of mirror is at distance of 2mm from point *P*. Point *P* and mirror are in the same plane of screen is perpendicular to this plane.



Q14. If fringe width is 0.1 mm then what is the wavelength of light used ?

(A) 3.3×10^{-7} m

(B) 6.7×10^{-7} m

(C) 1.0×10^{-7} m

(D) 4×10^{-7} m

Q15. If the mirror is shifted towards left then how does the fringe pattern on screen change?

(A) Fringe width decreases and the region in which interference is formed shifts downward.

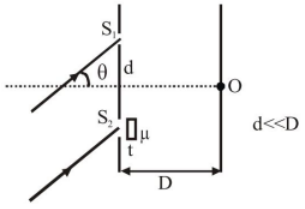
(B) Fringe width decreases and region in which inteference is formed shifts upwards.

(C) Fringe width does not change and region in which interference is formed shifts upwards.

(D) Fringe width does not change and region in which inteference is formed shifts downwards.

Paragraph for Question No. 16 to 18

A monochromatic beam of light falls on Young's double slit experiment apparatus as shown in figure. A thin sheet of glass is inserted in front of lower slit S_2.



Q16. The central bright fringe can be obtained

(A) at O only

(B) at O or below O only

(C) at O or above O only

(D) Anywhere on the screen

Q17. If central bright fringe is obtained on screen at *O* :-

(A) $(\mu-1)\, t = d\sin\theta$

(B) $(\mu-1)\, t = d\cos\theta$

(C) $(\mu-1)t + d\sin\theta = 0$

(D) $\dfrac{t}{\mu-1} = \dfrac{d}{\sin\theta}$

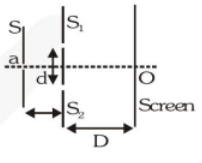
Q18. The phase difference between the waves interfering at fifth minima is :-

(A) 5π (B) 7π

(C) 9π (D) 11π

Paragraph for Question No. 19 to 21

The figure shows a schematic diagram showing the arrangement of Young's Double Slit Experiment :



Q19. Choose the correct statement(s) related to the wavelength of light used

(A) Larger the wavelength of light larger the fringe width.

(B) The position of central maxima depends on the wavelength of light used.

(C) If white light is used in YDSE, then the violet colour forms its first maxima closest to the central maxima.

(D) The central maxima of all the wavelengths coincide.

Q20. If the distance D is varied, then choose the correct statement(s)

(A) The angular fringe width does not change

(B) The fringe width changes in direct proportion

(C) The change in fringe width is same for all wavelengths

(D) The position of central maxima remains unchanged

Q21. If the distance d is varied, then identify the correct statement(s)

(A) The angular width does not change.

(B) The fringe width changes in inverse proportion.

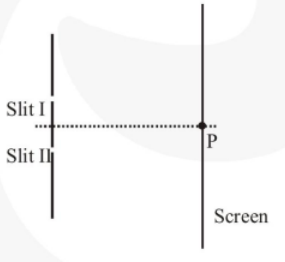
(C) The positions of all maxima change.

(D) The positions of all minima change.

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q22. A double slit interference pattern is produced on a screen, as shown in the figure, using monochromatic light of wavelength 500nm. Point P is the location of the central bright fringe, that is produced when light waves arrive in phase without any path difference. A choice of three strips A, B and C of transparent materials with different thicknesses and refractive indices is available, as shown in the table. These are placed over one or both of the slits, singularly or in conjunction, causing the interference pattern to be shifted across the screen from the original pattern. In the column–I, how the strips have been placed, is mentioned whereas in the column–II, order of the fringe at point P on the screen that will be produced due to the placement of the strips(s), is shown. Correctly match both the column.



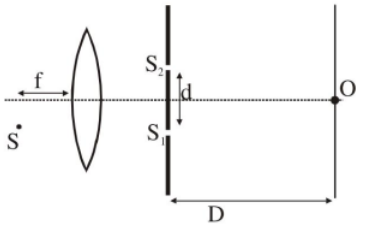
Film	A	B	C
Thickness (in μm)	5	1.5	0.25
Refractive index	1.5	2.5	2

Column I	**Column II**
(A) Only strip B is placed over slit–I	(P) First Bright
(B) Strip A is placed over slit–I and strip C is placed over slit–II	(Q) Fourth Dark
(C) Strip A is placed over the slit–I and strip B and strip C are placed over the slit–II in conjunction	(R) Fifth Dark
(D) Strip A and strip C are placed over slit–I (in conjuction) Bright and strip B is placed over Slit–II	(S) Central

Q23. Column-I shows some modifications in a standard YDSE setup. Column-II shows the associated characteristics.

Column-I **Column-II**

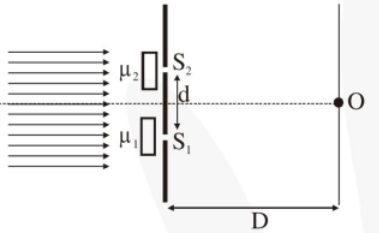
(A)



monochromatic point source S placed in focal plane.

(P) Zero order maxima lies above O.

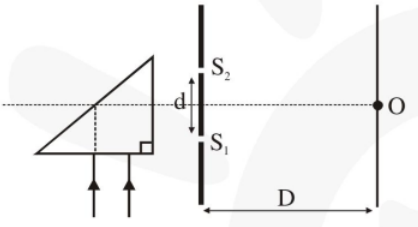
(B)



monochromatic parallel beam incident on S_1S_2 through transparent slabs of same thickness but $\mu_1 > \mu_2$

(Q) If a transparent mica sheet is introduced infront of S_2 central bright fringe can be obtained at O.

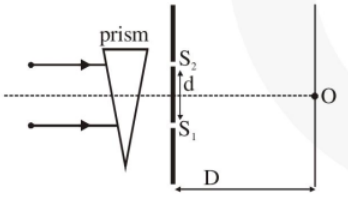
(C)



monochromatic parallel beam incident on a right angled isosceles prism of refractive index 1.50

(R) Fringe width $\beta = \dfrac{\lambda D}{d}$

(D)



monochromatic parallel beam incident on thin prism

(S) Point O can be a minima

(T) Point O can be a least order minima.

NUMERICAL TYPE

Q24. In young's double slit experiment, the slits are 2mm apart and are illuminated with a mixture of two wavelength, λ = 7500Å and λ' = 9000Å. At what minimum distance from the common central on a screen 2m from the slits will a bright fringe from one interference pattern coincide with a bright fringe from the other.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	A	A	D	D	A	D	A	D	D
11	12	13	14	15	16	17	18	19	20
A	D	A	C	C	C	D	B	B	D
21	22	23	24	25	26	27	28	29	30
C	D	A	C	B	B	B	D	D	B

Ex-1A

1. B,D **2.** B,C **3.** A,C,D **4.** B,C,D **5.** D

6. C **7.** B **8.** A **9.** B **10.** C

11. $\frac{K}{2}$ unit **12.** 1.76×10^{-3} m

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	D	C	C	B	B	B	D	D	D
11	12	13	14	15	16	17	18	19	20
D	D	D	A	A	A	A	B	A	C
21	22	23							
C	D	C							

Ex-2A

1. C **2.** A,C **3.** A,B **4.** C **5.** C

6. D **7.** C **8.** C **9.** A **10.** A,D

11. D **12.** A **13.** C **14.** B **15.** C

16. D **17.** A **18.** C **19.** A,C,D **20.** A,B,D

21. B,C,D

22. (A) → (R) ; (B) → (R) ; (C) → (S) ; (D) → (P)

23. (A) → (PRST); (B) → (QRST); (C) → (R); (D) → (PRST) **24.** 4.5 mm